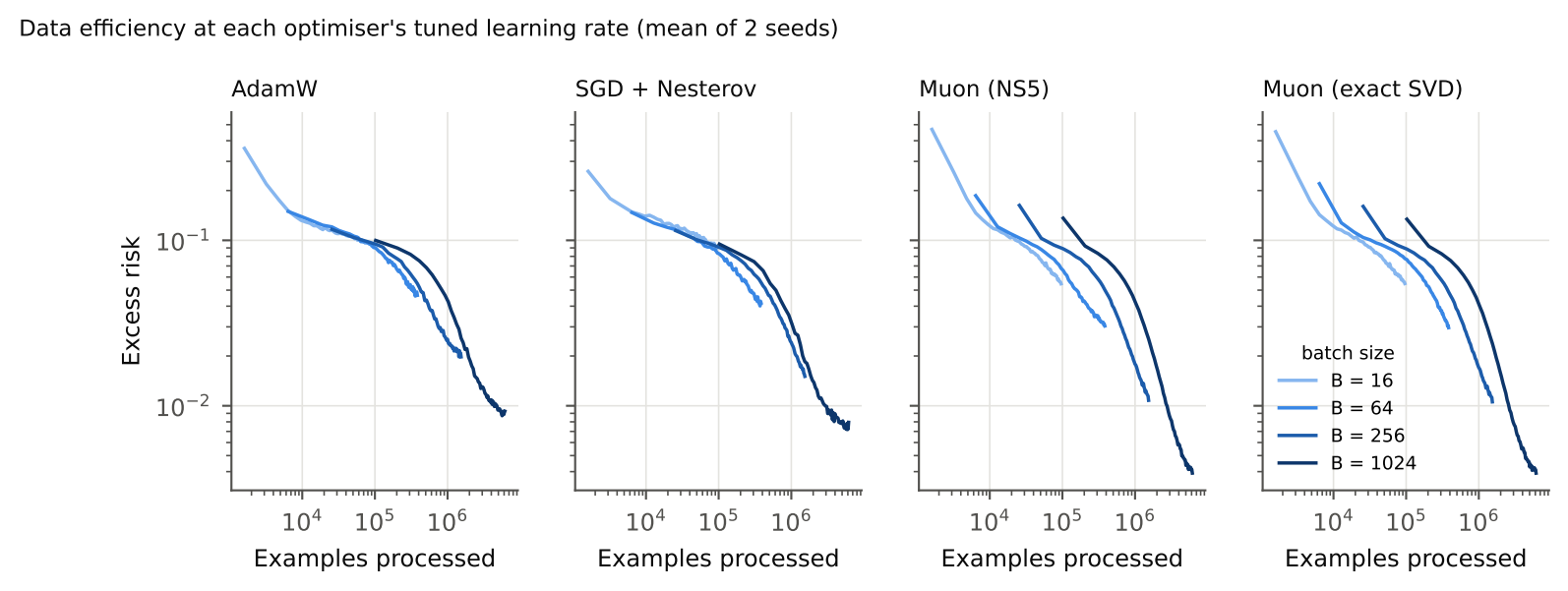
Data efficiency at each optimiser's tuned learning rate (mean of 2 seeds)
AdamW
SGD + Nesterov
Muon (NS5)
Muon (exact SVD)
Excess risk
$10^{-1}$
$10^{-2}$
$10^4$
$10^5$
$10^6$
Examples processed
batch size
B = 16
B = 64
B = 256
B = 1024